A Calibration Optimization Method for a Welding Robot Laser Vision System Based on Generative Adversarial Network

Yanbiao Zou, Jiaxin Chen, and Xianzhong Wei

***Abstract*—The accuracy and efficiency of a calibration process have a great influence on the robot locating accuracy. In the welding robot laser vision system discussed in this article, traditional structured light and hand–eye calibration methods are complex and time consuming. They require many manually collected labeled data, which leads to low calibration efficiency and accuracy. To solve the above problems, a calibration optimization method for the welding robot laser vision system based on the generative adversarial network is proposed in this article. By establishing an accurate calibration model, the calibration process is transformed into an optimization problem for the calibration parameters. An efficient and intelligent collection method is adopted to provide accurate, diverse, and sufficient calibration data for the calibration optimization method. The experimental results verify that the proposed method greatly improves the efficiency of calibration data collection. The reference point positioning experiment demonstrates that our method has low uncertainty in point positioning. The tracking experiments prove that our method keeps good accuracy and robustness in the tracking system, especially when the welding robot's pose changes greatly and the tracking trajectory is complex and changeable.**

***Index Terms*—Data collection, generative adversarial network (GAN), parameter optimization, robot calibration.**

I. Introduction

CONVENTIONAL welding robots mostly use the mode of "teaching–reproduction" for welding. However, this mode has disadvantages such as complicated operation, welding thermal deformation, and workpiece clamping error, giving rise to large welding errors. These problems can be solved by using the automatic welding seam tracking method based on a laser vision system [1]–[3]. This method uses a camera to collect the laser stripes' images that contain seam position information, thereby realizing seam tracking, as shown in Fig. 1.

Before adopting the welding robot laser vision system to obtain the seam position information, structured light calibration and hand–eye calibration should be carried out. During the process of robot calibration, the quality of the calibration result depends on the calibration data and calibration method. Therefore, developing a method for collecting accurate, diverse, and sufficient calibration data and simplifying the calibration method are the main goals of this article.

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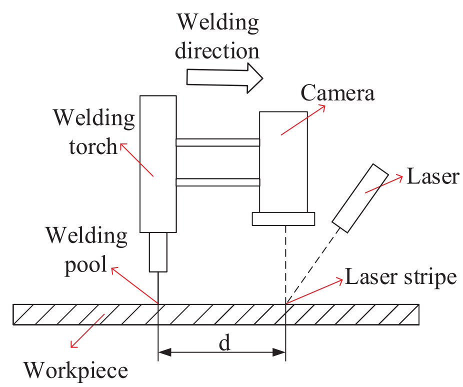

Fig. 1. Schematic of laser vision system.

A. Motivation

Calibration technology is widely used in sensor calibration and motion sensing [4]. There have been many studies on structured light calibration. McIvor [5] converted the calibration equation into a nonlinear least squares problem and used standard numerical methods to achieve structured light calibration. A new method proposed by Zhang and Huang [6] regarded the projector as a camera and can complete the structured light calibration of the projector accurately and quickly.

In terms of hand–eye calibration, Shiu and Ahmad [7] and Tsai *et al.* [8] pointed out that the problem of hand–eye calibration is actually to solve the equation $AX = XB$. A is the robot wrist motion, B is the resulting wrist-mounted sensor motion, and X is the conversion between the sensor coordinate system mounted on the robot wrist and the robot wrist coordinate system. Daniilidis [9] used the combination of dual-quaternion parameterization and singular value decomposition to solve this equation. Park and Martin [10] introduced the Lie theory to this equation and obtained a closed-form exact solution that could be visualized geometrically. Nguyen and Pham [11] strictly deduced the covariance of X. The experimental results demonstrated that their method could predict the covariance of hand–eye transformation accurately. Wu *et al.* [12] first proposed a unit-octonion representation to solve this equation. However, these methods are not applicable to the hand–eye calibration of the laser vision system.

Different from the general hand–eye calibration system, in our system, an optical filter is mounted in front of the camera to reduce the arc interference during welding. Due to the refraction of the optical filter, the results of general hand–eye calibration will be inaccurate after the installation of the optical filter. Thus, it is very important to design a calibration method suitable for the laser vision system.

In previous work, we proposed some hand–eye calibration optimization methods for a welding robot laser vision system [13], [14]. In [13], the ordinary least square (OLS) and semidefinite programming (SDP) methods were adopted to calculate the hand–eye calibration model. However, a long time was required to collect a large number of feature points, which led to low calibration efficiency. In [14], although the calibration method was simplified, it used the linear layers as the calibration model and did not consider that the rotation matrix of the hand–eye calibration must be orthogonal, greatly reducing the accuracy of the calibration model. In addition, the calibration data were all collected in the 2-D plane, while the calibration data in the 3-D space were not collected, giving rise to a problem of insufficient diversity of the calibration data.

As mentioned above, the traditional calibration methods have the following two shortcomings. First, they require many accurately labeled data. In the traditional collection method, on the one hand, to ensure the diversity and sufficiency of labeled data, a long time is required to collect sufficient data, greatly reducing the efficiency of calibration. On the other hand, the collection of labeled data is completed manually. This process is complex and tedious, making it difficult to guarantee the accuracy of the labeled data. It leads to low calibration accuracy. Second, structured light calibration is carried out before the hand–eye calibration, which is a continuous process. The errors produced by these two steps will accumulate and strongly affect the final calibration accuracy. Therefore, in order to improve the calibration accuracy and efficiency, it is highly important to ensure the accuracy, diversity, and sufficiency of the calibration data and simplify the calibration method.

Merz *et al.* [15] proposed semisupervised learning (SSL), which solved the problem of needing accurately labeled data for calibration. SSL adopted a large amount of unlabeled data as well as a small amount of labeled data for training, which achieved excellent results. There were many methods for SSL. In recent years, Goodfellow *et al.* [16] proposed the generative adversarial network (GAN), achieving excellent results in the field of SSL with its unique game idea. Odena [17] proposed semisupervised GAN (SGAN) and achieved good results on the MNIST dataset. SGAN not only significantly improved the quality of generated samples and the performance of classification but also reduced the training time. Ding *et al.* [18] proposed a new SSL method on the graph (GraphSGAN), which improved the Laplacian regularization of the traditional normalized graph. GraphSGAN obtained state-of-the-art results, obviously superior to the existing algorithms. Wei *et al.* [19] proposed a novel semisupervised deraining network (Semi-Deraingan) for a single image and successfully realized the rain removal. A large number of experimental results on public datasets showed that their model had excellent performance. GANs have a significant effect on SSL and provide a good idea for the training of calibration model.

B. Contributions

In this article, we propose a calibration optimization method for the welding robot laser vision system based on a GAN.

1) By establishing an accurate calibration model, the calibration process is transformed into an optimization problem for the calibration parameters. We use the calibration data to optimize the calibration parameters with the designed GAN. After that, a precise conversion from the pixel coordinate system to the robot base coordinate system can be got.
2) To provide accurate, diverse, and sufficient calibration data for the calibration optimization method, we propose an intelligent collection method. It collects a large amount of unlabeled calibration data and a small amount of labeled calibration data efficiently and intelligently.

The calibration data collection results show that the intelligent collection method can quickly and intelligently collect a large number of high-quality calibration data without too much manual participation, significantly improving the calibration efficiency. The reference point positioning experiment shows that the proposed method has low uncertainty and good stability in point positioning. The tracking experimental results demonstrate that the proposed method still has good accuracy and robustness when the welding robot's pose changes greatly and the tracking trajectory is complex and changeable.

The rest of this article is organized as follows. The structured light calibration model and hand–eye calibration model are built in Section II. The data collection method for calibration is designed in Section III. The GAN-based optimization method is proposed in Section IV. Experimental demonstrations are presented in Section V. The conclusions are summarized in Section VI.

II. Calibration Model

A. Structured Light Calibration Model

The perspective projection model is shown in Fig. 2. The pixel coordinates (c, r) of the weld feature point can be converted to camera coordinates (x_c, y_c, z_c) by (1). The details of the derivation have been thoroughly explained in our previous work [20]

$$\begin{cases} x_c = \dfrac{D_l(S_x c - C_x S_x)}{A_l(S_x c - C_x S_x) + B_l\left(S_y r - C_y S_y\right) + C_l f \xi} \\ y_c = \dfrac{D_l\left(S_y r - C_y S_y\right)}{A_l(S_x c - C_x S_x) + B_l\left(S_y r - C_y S_y\right) + C_l f \xi} \\ z_c = \dfrac{D_l f \xi}{A_l(S_x c - C_x S_x) + B_l\left(S_y r - C_y S_y\right) + C_l f \xi} \end{cases} \tag{1}$$

where $(S_x, S_y, f, K, C_x, C_y)$ refers to the intrinsic parameters of the camera and (A_l, B_l, C_l, D_l) are the parameters of the laser plane, and $\xi = K C_x^2 S_x^2 - 2K C_x S_x^2 c + K C_y^2 S_y^2 - 2K C_y S_y^2 r + K S_x^2 c^2 + K S_y^2 r^2 + 1$.

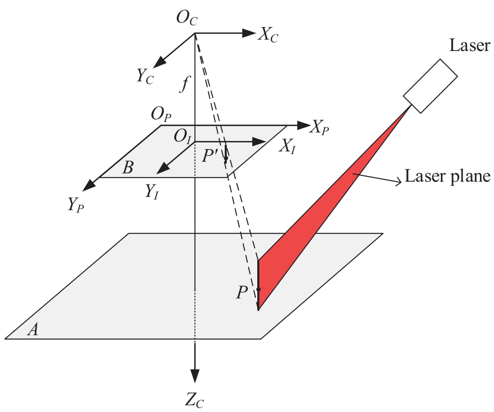

Fig. 2. Perspective projection model.

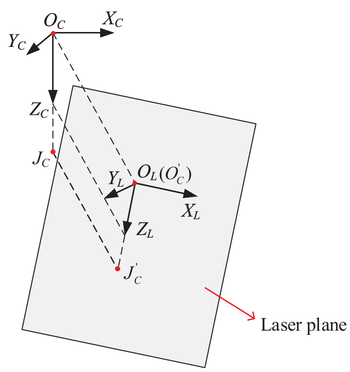

Fig. 3. Conversion from the 3-D coordinates of the camera to 3-D coordinates of laser plane.

Fig. 3 shows the conversion from the 3-D coordinates of the camera to the 3-D coordinates of the laser plane. Take any two initial points on the positive side of the Z-axis in the camera coordinate system. Here, we take the camera origin $O_C(0, 0, 0)$ and another point $J_C(0, 0, 100)$. J_C and O_C are projected onto the laser plane to obtain the projection points $J_C'(x_{jp}, y_{jp}, z_{jp})$ and $O_C'(x_{op}, y_{op}, z_{op})$. The projection formula of the point to the laser plane is as follows:

$$\begin{cases} x_p = x_0 - A_l t \\ y_p = y_0 - B_l t \\ z_p = z_0 - C_l t \end{cases} \tag{2}$$

where $t = A_l x_0 + B_l y_0 + C_l z_0 + 1/A_l{}^2 + B_l{}^2 + C_l{}^2$, (x_0, y_0, z_0) represents the coordinates of the initial point, and (x_p, y_p, z_p) represents the coordinates of the projected point.

The vector $\mathbf{Z}(x_{jp} - x_{op}, y_{jp} - y_{op}, z_{jp} - z_{op})$ is constructed from J_C' to O_C' and the unit vector $\mathbf{Z_L}$ is obtained by unitizing it. O_c' coincides with O_L, which is the origin of the laser plane coordinate system. The normal vector of the laser plane is $\mathbf{Y}(A_l, B_l, C_l)$, and the unit vector $\mathbf{Y_L}$ is obtained by unitizing it. The vector $\mathbf{X_L}$ is orthogonal to $\mathbf{Z_L}$ and $\mathbf{Y_L}$. The laser plane coordinate system is constructed by $\mathbf{X_L}$, $\mathbf{Y_L}$, and $\mathbf{Z_L}$. We, respectively, project three unit vectors $\mathbf{X_C}(1, 0, 0)$, $\mathbf{Y_C}(0, 1, 0)$, and $\mathbf{Z_C}(0, 0, 1)$ in the camera coordinate system to another three unit vectors $\mathbf{X_L}$, $\mathbf{Y_L}$, and $\mathbf{Z_L}$. We, respectively, project three unit vectors $\mathbf{X_C}(1, 0, 0)$, $\mathbf{Y_C}(0, 1, 0)$, and $\mathbf{Z_C}(0, 0, 1)$ in the camera coordinate system to another three unit vectors $\mathbf{X_L}$, $\mathbf{Y_L}$, and $\mathbf{Z_L}$ in the laser plane coordinate system. The calculation formula of the rotation matrix from the camera coordinate system to the laser plane coordinate system is as follows:

$${}_{\mathbf{C}}^{\mathbf{L}}\mathbf{R} = \begin{bmatrix} \mathbf{X_C} \cdot \mathbf{X_L} & \mathbf{Y_C} \cdot \mathbf{X_L} & \mathbf{Z_C} \cdot \mathbf{X_L} \\ \mathbf{X_C} \cdot \mathbf{Y_L} & \mathbf{Y_C} \cdot \mathbf{Y_L} & \mathbf{Z_C} \cdot \mathbf{Y_L} \\ \mathbf{X_C} \cdot \mathbf{Z_L} & \mathbf{Y_C} \cdot \mathbf{Z_L} & \mathbf{Z_C} \cdot \mathbf{Z_L} \end{bmatrix}. \tag{3}$$

The translation vector ${}_{\mathbf{C}}^{\mathbf{L}}\mathbf{T}(x_{op}, y_{op}, z_{op})$ is introduced because O_C translates (x_{op}, y_{op}, z_{op}) to obtain O_L. The transformation matrix ${}_{\mathbf{Camera}}^{\mathbf{Laser}}\mathbf{T}$ from the camera coordinate system to the laser plane coordinate system is obtained by combining equation (3)

$${}_{\mathbf{Camera}}^{\mathbf{Laser}}\mathbf{T} = \begin{bmatrix} {}_{\mathbf{C}}^{\mathbf{L}}\mathbf{R} & {}_{\mathbf{C}}^{\mathbf{L}}\mathbf{T} \\ 0 & 1 \end{bmatrix}. \tag{4}$$

The conversion from the camera coordinates (x_c, y_c, z_c) to the laser plane coordinates $(x_l, 0, z_l)$ can be achieved by using the following equation:

$$\begin{bmatrix} x_l \\ 0 \\ z_l \\ 1 \end{bmatrix} = {}_{\mathbf{Camera}}^{\mathbf{Laser}}\mathbf{T} \times \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}. \tag{5}$$

Thus, (c, r) can be used in (1)–(5) to calculate $(x_l, 0, z_l)$ to achieve structured light calibration.

Clearly, in the structured light calibration model, the camera's intrinsic parameters (S_x, S_y, f, K, C_x, and C_y) and laser plane parameters (A_l, B_l, C_l, and D_l) need to be determined. The accuracy of these parameters determines the accuracy of the structured light calibration model. The traditional calibration accuracy of the camera's intrinsic parameters is relatively high, so in this article, we adopted Heikkila's method [21] to calculate them. However, the determination of laser plane parameter (A_l, B_l, C_l, D_l) is still an urgent problem to be solved.

B. Hand-Eye Calibration Model

Hand–eye calibration is generally used to establish the conversion from the sensor coordinate system to the robot wrist coordinate system. In the welding robot based on the laser vision system, hand–eye calibration refers to the conversion from the laser plane coordinate system to the robot wrist coordinate system. The relative position of each coordinate system of the arc welding robot is shown in Fig. 4. The 3-D coordinate system $O_C X_C Y_C Z_C$ is the camera coordinate system, $O_L X_L Y_L Z_L$ is the laser plane coordinate system, $O_W X_W Y_W Z_W$ is the robot wrist coordinate system, $O_B X_B Y_B Z_B$ is the robot base coordinate system, and $O_T X_T Y_T Z_T$ is the robot tool coordinate system.

We assumed that P'' is a point in the robot workspace. The coordinate of P'' under the laser plane coordinate system is $(x_l, 0, z_l)$, and the coordinate under the robot wrist coordinate system is (x_w, y_w, z_w). ${}_{\mathbf{Laser}}^{\mathbf{Wrist}}\mathbf{T}$ is the transformation matrix from the laser plane coordinate system to the robot wrist

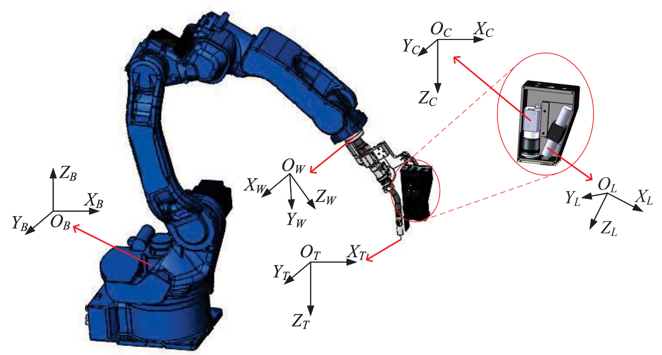

Fig. 4. Relative position of each coordinate system of arc welding robot.

coordinate system, and ${}^{\mathbf{Wrist}}_{\mathbf{Laser}}\mathbf{T}$ can be expressed as follows:

$$
{}^{\mathbf{Wrist}}_{\mathbf{Laser}}\mathbf{T} = \begin{bmatrix} {}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R} & {}^{\mathbf{W}}_{\mathbf{L}}\mathbf{T} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & dx \\ r_{21} & r_{22} & r_{23} & dy \\ r_{31} & r_{32} & r_{33} & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}
$$

where ${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R}$ and ${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{T}$ represent the rotation matrix and translation vector from the laser plane coordinate system to the robot wrist coordinate system, respectively. Therefore, their conversion is as follows:

$$
\begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} = {}^{\mathbf{Wrist}}_{\mathbf{Laser}}\mathbf{T} \times \begin{bmatrix} x_l \\ 0 \\ z_l \\ 1 \end{bmatrix}. \tag{7}
$$

There is orthogonality between column vectors of rotation matrix ${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R}$ based on its physical meaning, that is, $\mathbf{a_1} = [r_{11}, r_{21}, r_{31}]^T$, $\mathbf{a_2} = [r_{12}, r_{22}, r_{32}]^T$, and $\mathbf{a_3} = [r_{13}, r_{23}, r_{33}]^T$ are required to be orthogonal to each other. Given that $y_l = 0$, (7) can be simplified as

$$
\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} r_{11} & r_{13} & dx \\ r_{21} & r_{23} & dy \\ r_{31} & r_{33} & dz \end{bmatrix} \times \begin{bmatrix} x_l \\ z_l \\ 1 \end{bmatrix}. \tag{8}
$$

${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R}$ can be obtained by using the rotation angle Rx, Ry, and Rz. The specific calculation equation is as follows:

$$
\begin{cases} r_{11} = \cos(Rz)\cos(Ry) \\ r_{13} = \cos(Rz)\sin(Ry)\cos(Rx) + \sin(Rz)\sin(Rx) \\ r_{21} = \sin(Rz)\cos(Ry) \\ r_{23} = \sin(Rz)\sin(Ry)\cos(Rx) - \cos(Rz)\sin(Rx) \\ r_{31} = -\sin(Ry) \\ r_{33} = \cos(Ry)\cos(Rx). \end{cases} \tag{9}
$$

Since $\mathbf{a_1}$, $\mathbf{a_2}$, and $\mathbf{a_3}$ are orthogonal to each other, ${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R}$ obtained by the rotation angle Rx, Ry, and Rz is also orthogonal. Therefore, the orthogonality of the rotation matrix ${}^{\mathbf{W}}_{\mathbf{L}}\mathbf{R}$ is guaranteed, which further improves the accuracy of hand–eye calibration. It can be seen from (8) and (9) that if accurate rotation angle Rx, Ry, and Rz and the translation vector dx, dy, and dz are obtained, the conversion between the laser plane coordinate system and the robot wrist coordinate system can be established. Then, hand–eye calibration can be realized.

In the welding robot system, the conversion ${}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T}$ between the robot wrist coordinate system and the robot base coordinate system is obtained by the calibration algorithm in the robot controller.

Therefore, the coordinates of P'' in the robot base coordinate system can be obtained to prepare for the subsequent design of the GAN model and tracking experiment. The calculation formula is as follows:

$$
\begin{bmatrix} x_b \\ y_b \\ z_b \\ 1 \end{bmatrix} = {}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T} \times \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} + \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \\ 0 \end{bmatrix} \tag{10}
$$

where Δx, Δy, and Δz are model correction terms. Due to the uncertainty of the parameters in the robot kinematics, there is a position error at the robot end. To reduce the systematic effects, it is necessary to add Δx, Δy, and Δz for correction and compensation to make robot movement more accurate. In [13] and [14], Δx, Δy, and Δz are designed manually based on experience, with low reliability and accuracy. The model correction terms Δx, Δy, and Δz at the robot end are related to the pose of the robot wrist. Its calculation equation of is as follows:

$$
\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \mathbf{T_{Pose}} \begin{bmatrix} Rx_{\text{wrist}} \\ Ry_{\text{wrist}} \\ Rz_{\text{wrist}} \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} Rx_{\text{wrist}} \\ Ry_{\text{wrist}} \\ Rz_{\text{wrist}} \end{bmatrix} \tag{11}
$$

where $[\,Rx_{\text{wrist}}\ Ry_{\text{wrist}}\ Rz_{\text{wrist}}\,]^T$ are the pose of the robot wrist and $\mathbf{T_{Pose}}$ is a coefficient matrix between $[\,Rx_{\text{wrist}}\ \ Ry_{\text{wrist}}\ \ Rz_{\text{wrist}}\,]^T$ and $[\,\Delta x\ \ \Delta y\ \ \Delta z\,]^T$. $\mathbf{T_{Pose}}$ can be obtained through massive data optimization. $[\,\Delta x\ \ \Delta y\ \ \Delta z\,]^T$ can be changed with the robot's attitude, so the positioning error of robot end can be better corrected.

Therefore, by combining (1), (5), (8), and (10), we can use the pixel coordinate (c, r) of a certain point on the laser stripe to calculate its coordinate (x_b, y_b, z_b) in the robot base coordinate system.

It can be seen from Sections II-A and II-B that to establish an accurate calibration model and improve calibration accuracy, the calibration parameters must be optimized. In this article, the calibration parameters are $(A_l, B_l, C_l, D_l, Rx, Ry, Rz, dx, dy, dz$, and $\mathbf{T_{Pose}})$. Therefore, the traditional calibration problem is transformed into the parameter optimization problem. The accuracy of calibration can be improved by obtaining accurate calibration parameters.

III. Calibration Data Collection

It can be seen from Section I that the calibration accuracy depends on the accurate, diverse, and sufficient calibration data. Therefore, the method used for the collection of the calibration data is essential. We adopt an efficient and intelligent collection method that collects a large amount of unlabeled calibration data and a small amount of labeled calibration data by scanning an iron plate with the multipose laser vision system.

Ten groups of calibration data are collected for the iron plate that was placed on the workpiece platform in different positions and poses. The iron plate placement distribution covered most of the robot's movement range, which made the

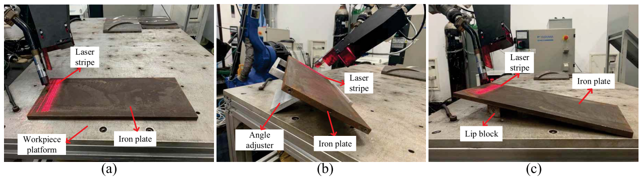

Fig. 5. Distribution of three groups of iron plate: (a) iron plate is placed on the workpiece platform, (b) iron plate is placed on the angle adjuster, and (c) iron plate is placed on the lip block.

calibration data more diverse and reliable. Fig. 5 shows the distribution of three iron plate groups. During pose adjustment, the iron plate was placed on the angle adjuster or lip block. We ensured the diversity of calibration data by changing the angle of the iron plate. Therefore, compared with [14], this method not only collected the 2-D calibration data but also collected the 3-D one, greatly improving the diversity of the intelligent calibration data and preparing high-quality calibration data for the subsequent 3-D tracking of the welding robot. During the collection of each group of calibration data, the welding torch end of the welding robot was aligned with noncollinear three points of the iron plate plane (including a teaching starting point, a teaching ending point, and a third point). The coordinate values of three points in the robot base coordinate system were obtained from the robot teaching pendant to calculate the equation of the iron plate plane. Because the coordinates of these three points accurately determined the position and pose of the iron plate, the coordinate data of the three points were taken as part of the labeled data. The labeled data were easily and accurately obtained, ensuring the accuracy of the calibration data. Then, we taught the robot to move from the teaching starting point to the teaching ending point with changing poses. At the same time, the camera captured images and the robot controller calculated ${}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T}$ in real time. In the data collection process, the robot's pose changed greatly, further improving the diversity of the calibration data. In the end, we used the line morphological method to extract the pixel coordinates of the feature point in the images. In each group, we obtained three points under the robot base coordinate system as labeled data and approximately 17 000 pixel coordinates and ${}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T}$ as unlabeled data, solving the problem of the insufficient amount of calibration data. The process of extracting pixel coordinates of the feature point by the line morphological method is shown in Fig. 6.

To evaluate our method more comprehensively, we compared it with the traditional methods in [8], [10]–[12], and [14]. We manually operated the robot and changed various poses to photograph the calibration pattern; 279 high-quality calibration images were collected by Halcon Software, requiring approximately hours. Due to the limitations of manual operation, the accuracy of the calibration images is difficult to guarantee. For comparison, our method only needed to manually locate the position and pose of three points, without excessive manual operation. Moreover, we collected 17 151 calibration images in only one hour, further highlighting the high efficiency and intelligence of our collection method. Thus, our method not only guaranteed the sufficiency,

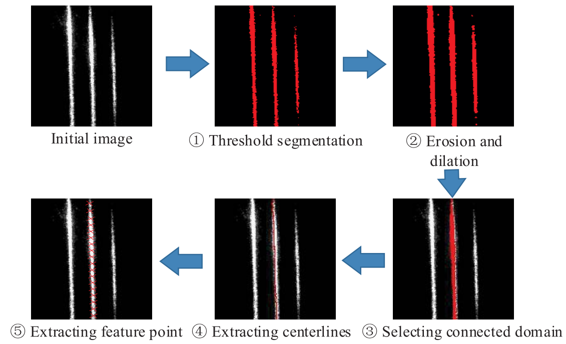

Fig. 6. Process of extracting pixel coordinates of the feature point by the line morphological method.

diversity, and accuracy of calibration data but also provided a good data source for the subsequent training and testing of the GAN model.

IV. Calibration Algorithm

A. Principle of GAN

Goodfellow *et al.* [16] proposed a GANs framework, including an adversarial network D and a generative network G. This framework is equivalent to a minimax two-player game. G is responsible for learning the distribution of original data, and D is responsible for discriminating whether a sample comes from the training data or the data generated by G. The goal of G is to learn the distribution of real data and the goal of D is to improve the discriminating ability and distinguish the generated data from the real data. D and G are optimized by solving the minimax problem as follows:

$$\min_G \max_D V(D, G) = E_{x\sim p_{\text{data}}(x)}\big[\log D(x)\big] + E_{z\sim p_z(z)}\big[\log(1 - D(G(z)))\big] \quad (12)$$

where $E(\cdot)$ is the expectation operator, x and z are the training data and sample noise, respectively, and $P_{\text{data}}(x)$ and $P_z(z)$ represent the probability distributions of the training data and sample noise, respectively. The sample generated by G is defined as $G(z)$. $D(x)$ and $D(G(z))$ represent the discriminant probability of input from the training data and the generated samples, respectively. G attempts to minimize $\log(1 - D(G(z)))$, that is, to make $V(D, G)$ as small as possible. By contrast, D attempts to maximize $D(x)$ and minimize $D(G(z))$ at the same time. In the training process, D and G play against each other and finally reach equilibrium.

However, due to the problem of vanishing gradient, the training of GANs is not easy to carry out. As an improvement, the use of batch normalization in both generator and discriminator had been found to be beneficial when gradients were lost or exploded. Zhang *et al.* [22] proposed text GAN (textGAN) and successfully applied it to discrete natural language processing. Experiments showed that the textGAN generated real sentences after adversarial training. Inspired by [22], this article designs a GAN combined with the calibration model of the welding robot and optimizes the calibration parameters by SSL.

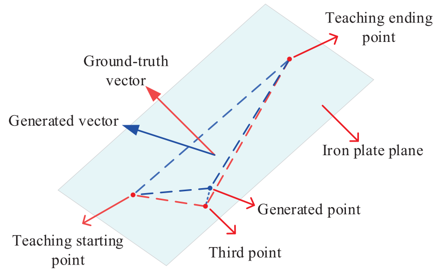

Fig. 7. Method to calculate the ground-truth vector and generated vector.

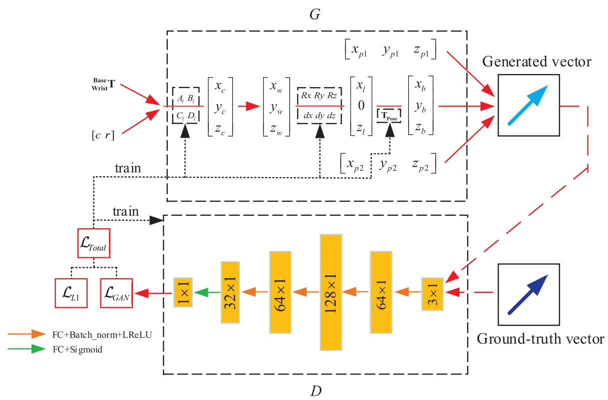

Fig. 8. Network architecture of GAN. Note: $[x_{p1}\ y_{p1}\ z_{p1}]$ and $[x_{p2}\ y_{p2}\ z_{p2}]$ represent the coordinates of teaching starting point and ending point in the robot base coordinate system.

B. Construction of the GAN Model

In Section III, we obtained a small amount of labeled data and a large amount of unlabeled data. The exact iron plate plane equation was obtained from the labeled data. In this article, the iron plate plane was formed by a teaching starting point, a teaching ending point, and a third point. By calculating the normal vector of this plane, we obtained the ground-truth vector. The unlabeled data (pixel coordinates (c, r) and ${}^{\text{Base}}_{\text{Wrist}}\mathbf{T}$) can be used to generate the corresponding robot base coordinates, namely the generated points, through the calibration model in Section II. The generated point, teaching starting point, and teaching ending point formed a generated plane, and the normal vector of the plane was calculated as the generated vector. The points extracted by the line morphology method in Section III were theoretically on the iron plate plane. With more accurate calibration parameters, the distance between the points generated by the calibration model and the iron plate plane should tend to mm. Similarly, the generated plane and the iron plate plane tend to coincide, and the generated vector and ground-truth vector tend to coincide. As shown in Fig. 7, we used the difference between the generated vector and the ground-truth vector to measure the distance between the generated point and the iron plate plane. The smaller difference between the generated vector and the ground-truth vector, the higher the calibration accuracy. Therefore, we designed a GAN model suitable for the welding robot laser vision system, as shown in Fig. 8.

Similar to traditional GANs, our method also contains two structures: a discriminative network D and a generative model G. In G, its structure adopts the calibration model described in Section II. From the input pixel coordinates (c, r) and the corresponding conversion ${}^{\text{Base}}_{\text{Wrist}}\mathbf{T}$, G outputs the generated vector. Five fully connected layers are used to construct discriminator D. For the first four fully connected layers (i.e., 64×1, 128×1, 64×1, and 32×1 in Fig. 8), batch normalization and leaky rectified linear (LeakyReLU) are applied to each of them. We use the sigmoid function to activate the last fully connected layer. D and G are used in the training process, and in the inference process, only G is used. In the training process, the generated vector and the ground-truth vector are input into D, and the loss $\mathcal{L}_{\text{GAN}}$ is output. The loss function $\mathcal{L}_{\text{GAN}}$ is calculated as follows:

$$\mathcal{L}_{\text{GAN}}(D, G) = E_x\left[\log D(x)\right] + E_z\left[\log(1 - D(G(z)))\right] \quad (13)$$

where x represents ground-truth vector and z represents the input of G that is pixel coordinates (c, r) and corresponding conversion ${}^{\text{Base}}_{\text{Wrist}}\mathbf{T}$. $G(z)$ represents generated vector.

To facilitate better convergence, the $L1$ loss $\mathcal{L}_{L1}(G)$ is introduced, representing the distance from the generated points to the iron plate plane

$$\mathcal{L}_{L1}(G) = \frac{\left|A_g x_b + B_g y_b + C_g z_b + D_g\right|}{\sqrt{A_g^2 + B_g^2 + C_g^2}} \quad (14)$$

where A_g, B_g, C_g, and D_g represent the coefficients of the iron plate plane equation. Therefore, the total loss of the GAN model can be expressed as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{GAN}}(G, D) + \lambda \mathcal{L}_{L1}(G) \quad (15)$$

where λ is the weight of $\mathcal{L}_{L1}(G)$. Thus, the ultimate optimization goal is

$$G^* = \arg\min_G \max_D \mathcal{L}_{\text{Total}}. \quad (16)$$

Finally, $\mathcal{L}_{\text{Total}}$ is used to train the parameters of D and the calibration parameters in G. G and D play games against each other, so D cannot distinguish the difference between the generated vectors and the ground-truth vectors. Similarly, the distance from the generated points to the iron plate plane $\mathcal{L}_{L1}(G)$ is also minimized, and the optimized calibration parameter results are obtained.

C. Training and Testing

1) Training the GAN Model: Seven groups of intelligent calibration data were selected as the training set, and three groups of intelligent calibration data were selected as the test set. The parameters of G and D were trained under the Pytorch framework. Glorot's method [23] was used to initialize the weight parameters of D. The initial learning rates of G and D were set to 0.001. The weight of the $L1$ loss λ in (15) was set to 0.01. The minibatches of G and D were set to 200. We randomly input the training data (x_i, z_i) obtained from Section III into the GAN model for training. x_i represents the ground-truth vector, and z_i represents the pixel coordinates and corresponding conversion ${}^{\text{Base}}_{\text{Wrist}}\mathbf{T}$. The Adam optimizer was adopted to optimize the parameters of D and G through the gradient descent method. Experience showed that in the

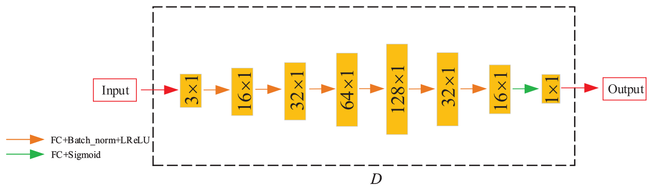

Fig. 9. Structure of the discriminator with more complex network architecture.

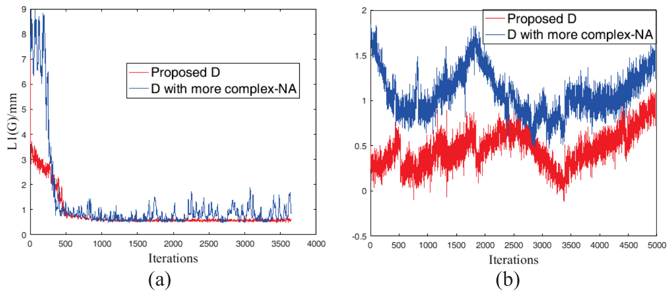

Fig. 10. Comparison experiment of the calibration results: (a) training result and (b) testing result.

process of training the network, the total loss $\mathcal{L}_{\text{Total}}$ continued to decrease and converged quickly. The number of training iteration was set to 50.

2) Testing the GAN Model: The optimized calibration parameters $(A_l, B_l, C_l, D_l, Rx, Ry, Rz, dx, dy, dz,$ and $\mathbf{T}_{\mathbf{Pose}})$ were obtained after training. The data from the test set are used to verify the accuracy of calibration parameters. The calculated $\mathcal{L}_{L1}(G)$ value is 0.5 mm, which meets the requirements of engineering application.

D. Ablation Study on Verifying the GAN Architecture Design

In order to further verify the effectiveness of our GAN architecture design, we conducted two ablation studies.

1) Ablation Study of Discriminators With Different Network Architectures: In this ablation study, we ensure that the discriminators have different architectures and similar complexity. We designed a discriminator with more complex network architecture that consists of eight full connected layers, as shown in Fig. 9. We used the training set and test set consistent with the proposed method to train and verify this GAN model. $\mathcal{L}_{L1}(G)$ is used to evaluate the training accuracy and test accuracy. The results are shown in Fig. 10. The mean absolute error (MAE) and the standard deviation (SD) of $\mathcal{L}_{L1}(G)$ are shown in Table I.

As shown in Fig. 10(a), during the training process, the proposed D converges faster and reaches a smaller error than that with more complex network architecture. It can be seen from Fig. 10(b) and Table I that the proposed D is more accurate and stable than the more complex one in the test set. The experimental results show that when the discriminators have the same complexity level but with different architectures, the discriminator with too complex architecture does not discriminate between the generated and the ground-truth vectors well.

TABLE I
TEST ERRORS OF TWO METHODS

Method	MAE(mm)	SD(mm)
D with more complex-NA	1.08	0.44
Proposed D	0.47	0.31

[1] Note: Proposed D and D with more complex-NA denote the proposed discriminator and the discriminator with more complex network architecture, respectively.

2) Ablation Study of Discriminators With Different Complexity Levels in the Same Architecture: In this ablation study, we ensure that the discriminators have different complexity levels and the same architecture, which all consists of five fully connected layers but with a different number of parameters. The parameters of each layer of different models are shown in Table II. Similarly, we conducted a series of comparative experiments. The MAE and SD of $\mathcal{L}_{L1}(G)$ are shown in Table III.

It can be seen from Table III that compared with the proposed D, the discriminators with lower or higher complexity have higher MAE and SD. By comparing the performance between different complexity levels of the same architecture for the discriminator, the proposed GAN model is more suitable for the calibration problem.

V. EXPERIMENTS

The movement accuracy of the robot is highly important, particularly in the environment where the robot's pose changes continuously and tracks complex 2-D and 3-D curves. Therefore, it is very important to guarantee the accuracy of robot calibration. Therefore, to test the correctness of our method, in Section V, the reference point positioning experiment, the angle iron tracking experiment with continuous large pose change, and the welding tracking experiment with complex 2-D and 3-D curves are carried out.

A. Experimental Platform

The hardware of the calibration experiment platform mainly includes an execution mechanism, a laser vision system, and an industrial control computer, as shown in Fig. 11. The execution mechanism adopted a six-axis welding manipulator (model YASKAWA MA1440), which has a repeatable positioning accuracy of ±0.08 mm. The main specifications of the laser vision system are listed in Table IV. The configuration of the industrial control computer is an IPC-710 industrial computer with an Intel i7-3770 CPU, 16-GB RAM, and NVIDIA GeForce GTX-1080 GPU.

The connections between the parts of the automatic seam tracking system are shown in Fig. 12. The connections between the parts of the automatic seam tracking system are shown in Fig. 12. The laser vision sensor collects the welding images and transmits them to the industrial computer via Ethernet during the welding process. The industrial computer calculates the position information by an algorithm and transmits it to the robot control cabinet. Then, the robot control

TABLE II
PARAMETERS OF EACH LAYER OF DIFFERENT MODELS

The name of the discriminator	Lower-1 D	Lower-2 D	Proposed D	Higher-1 D	Higher-2 D
Layer 1	16	32	64	64	64
Layer 2	32	64	128	256	256
Layer 3	16	32	64	64	128
Layer 4	8	16	32	32	64
Layer 5	1	1	1	1	1

[1] Note: Lower-N D represents the N^{th} lower-complexity discriminator and Higher-N D represents the N^{th} higher-complexity discriminator.

TABLE III
TEST ERRORS OF FIVE METHODS

Method	Lower-1 D	Lower-2 D	Proposed D	Higher-1 D	Higher-2 D
MAE(mm)	1.5	0.9	0.47	1.08	1.95
SD(mm)	0.66	0.57	0.31	0.58	0.73

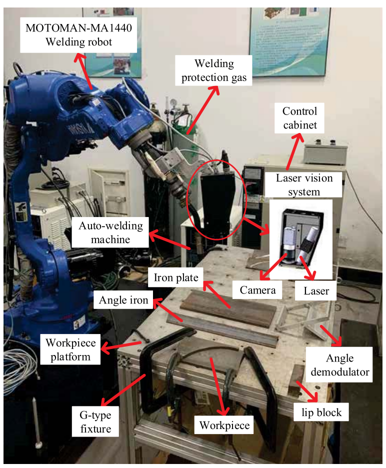

Fig. 11. Welding experiment platform.

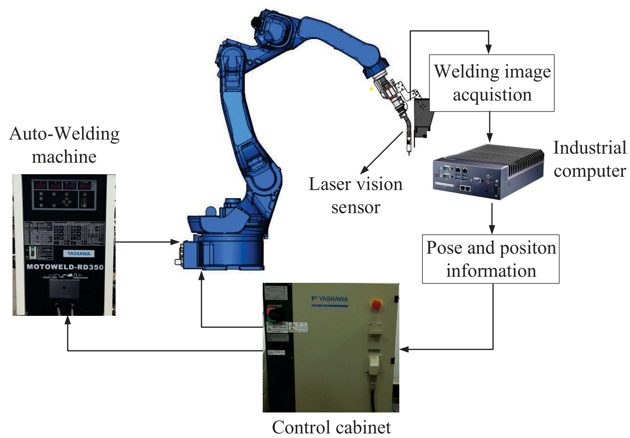

Fig. 12. Connections between the parts of the automatic seam tracking system.

TABLE IV
LASER VISION SYSTEM PARAMETERS

Parameter	Value
Diode power of laser	100mW
Wavelength of laser	660nm
Intensity of laser	Uniform lengthwise
Distribution of laser	Gaussian widthwise
Fan angles of laser	30°
center wavelength of optical filter	660 ± 10nm
Resolution of camera ($V \times H$)	$1280 \times 1024 pixel^2$
Field of view of camera	$160 \times 128 mm^2$
Scale factor of camera	$0.125 mm/pixel$
sampling rate of camera	60fps
Interface of camera	GigE

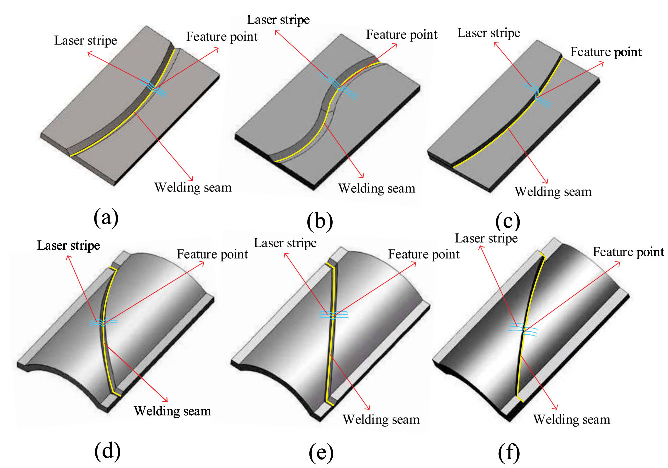

Fig. 13. Structure diagram of six welding workpieces. (a) 2-D-B/C. (b) 2-D-L/C. (c) 2-D-L/S. (d) 3-D-B/C. (e) 3-D-B/I. (f) 3-D-L/I.

cabinet generates a signal and sends it to the welding machine. After that, the welding machine transmits the signal to the welding robot to control the movement of the welding robot. Six types of welding workpieces are shown in Fig. 13. Its specifications are listed in Table V.

B. Experiment of Feature Point Position

It can be seen from Section II that before obtaining the robot base coordinates of the tracking feature points, we should extract the pixel coordinates (c, r) of the feature points of the workpieces as the initial input. In this experiment, we used the point morphology method [24] to extract the

TABLE V
SPECIFICATIONS OF WELDING WORKPIECES

Type	Specifications
2D-B/C	2D, butt joint, C-shaped weld seam
2D-B/S	2D, butt joint, S-shaped weld seam
2D-L/C	2D, lap joint, C-shaped weld seam
3D-B/C	3D, butt joint, C-shaped weld seam
3D-B/I	3D, butt joint, I-shaped weld seam
3D-L/I	3D, lap joint, I-shaped weld seam

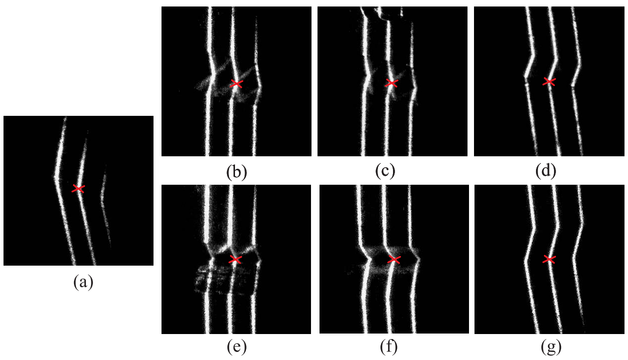

Fig. 14. Extraction of feature points of various workpieces. (a) Angle iron. (b) 2-D-B-C. (c) 2-D-B-S. (d) 2-D-L-C. (e) 3-D-B-C. (f) 3-D-B-I. (g) 3-D-L-I. Note: red cross represents the extracted feature point.

feature points. The extraction results of various workpieces are shown in Fig. 14.

To ensure the accuracy of the extracted feature points, we collected two groups of tracking images for each workpiece and created labeled feature points. Then, the point morphology method was adopted to extract the pixel coordinates of the feature points of the tracking images. The Euclidean distance between the feature points extracted by the point morphology method and the labeled feature points is used to calculate the position errors of the feature points. Its calculation formula is as follows:

$$E_i = \sqrt{\left(x_i^* - x_i\right)^2 + \left(y_i^* - y_i\right)^2} \tag{17}$$

where i denotes the sequence number of the image frame, (x_i, y_i) denotes the coordinates of the extracted feature point, and (x_i^*, y_i^*) denotes the coordinates of labeled feature point. To better measure the experimental results, this experiment calculates the SD and MAE of each group of errors. We adopted the MAE and SD to evaluate the accuracy and stability of the method, respectively. The position errors of each workpieces are shown in Table VI. The number of frames per group of the tracking images is expressed by the sequence length.

The maximum width of the laser stripe is 10 pixels. As shown in Table VI, the point morphology method can keep the MAE of the feature point position within 5 pixels and the SD within 3 pixels. Therefore, the point morphology method can stably and accurately extract the pixel coordinates of feature points and provide accurate input for the subsequent calibration model.

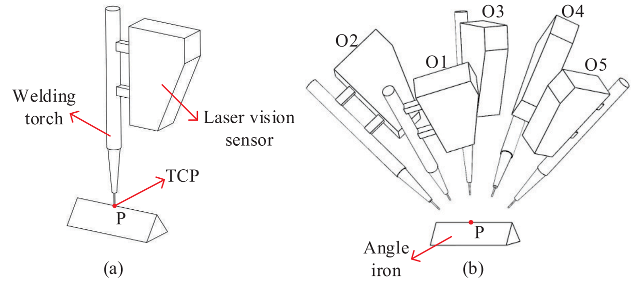

Fig. 15. Obtain the position of the reference point. (a) Obtain the position by torch center point (TCP). (b) Collect the images of the reference point by using the laser vision system from different positions and poses.

C. Experiment of Reference Point Positioning

To verify the calibration accuracy of the proposed method for a certain reference point under different robot positions, we used angle iron as the experimental workpiece. The welding torch was aligned with a reference point, and the coordinate values of this point in the robot base coordinate system (x_g, y_g, z_g) were obtained through the robot teaching pendant. Then, the laser vision system was used to collect the image of reference point from ten different positions and poses (O1–O10). At the same time, ${}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T}$ corresponding to each image was recorded. The collection process of five groups of reference point positions is shown in Fig. 15. The pixel coordinates $(c_i, r_i)\{i \in N|1 \le i \le 10\}$ of the reference points were extracted by the point morphology method of Section V-B. The proposed method was adopted to calculate the coordinates of the reference point $(x_{pi}, y_{pi}, z_{pi})\{i \in N|1 \le i \le 10\}$ in the robot base coordinate system.

There is uncertainty in all measurements, but the width of the laser stripe is the main reason for the uncertainty of this experiment. The uncertainty exists in obtaining the pixel coordinates of the reference points. Here, the standard uncertainty μ_c, μ_r of (c_i, r_i) was calculated by type A evaluation of standard uncertainty [25]

$$\mu_v = \sqrt{\frac{\sum_{i=1}^{n}\left(v_i - v_g\right)^2}{n(n-1)}} \tag{18}$$

where n is the total number of reference points, v_i represents the pixel coordinates (c_i, r_i) of the reference points, and v_g represents the labeled reference point in the pixel coordinate system. After obtaining μ_v, the standard uncertainties μ_{vx}, μ_{vy}, and μ_{vz} of μ_v relative to the three directions of the robot base coordinate system can be calculated according to the following equation:

$$\begin{cases} \mu_{xv} = \left|\dfrac{\partial x}{\partial v}\right| \mu_v \\ \mu_{yv} = \left|\dfrac{\partial y}{\partial v}\right| \mu_v \\ \mu_{zv} = \left|\dfrac{\partial z}{\partial v}\right| \mu_v. \end{cases} \tag{19}$$

Assuming that $[\,\mu_{cx} \;\; \mu_{cy} \;\; \mu_{cz}\,]$ and $[\,\mu_{rx} \;\; \mu_{ry} \;\; \mu_{rz}\,]$ correspond to the uncertainties of c_i and r_i, respectively, the combined standard uncertainty $[\,\mu_{Cx} \;\; \mu_{Cy} \;\; \mu_{Cz}\,]$ is

TABLE VI
POSITION ERRORS OF THE DATA

Workpieces types	Angle iron		2D-B/C		2D-B/S		2D-L/C		3D-B/C		3D-B/I		3D-L/I	
Group number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Sequence length (frames)	852	841	677	662	692	679	681	701	603	602	593	597	565	575
MAE (pixels)	3.54	3	2.67	2.71	2.95	3.21	3.4	4.37	3.84	4.8	4.38	3.94	2.76	3.2
SD (pixels)	1.6	1.21	1.53	1.85	1.76	1.77	1.33	2	1.44	2.85	1.71	2.32	1.19	1.56

TABLE VII
EXPERIMENTAL RESULTS OF REFERENCE POINT POSITIONING

(x_g, y_g, z_g)	(x_{p1}, y_{p1}, z_{p1})	(x_{p2}, y_{p2}, z_{p2})	(x_{p3}, y_{p3}, z_{p3})	(x_{p4}, y_{p4}, z_{p4})	(x_{p5}, y_{p5}, z_{p5})	(x_p, y_p, z_p)	Δe
(1143.77, -477.44, 35.27)	(1144.01, -477.57, 34.36)	(1143.76, -477.99, 35.13)	(1144.06, -477.68, 34.93)	(1144.07, -477.77, 35.35)	(1144.47, -476.83, 34.84)	(1143.84±0.41, -477.65±0.33, 34.95±0.53)	0.39
	(x_{p6}, y_{p6}, z_{p6})	(x_{p7}, y_{p7}, z_{p7})	(x_{p8}, y_{p8}, z_{p8})	(x_{p9}, y_{p9}, z_{p9})	$(x_{p10}, y_{p10}, z_{p10})$		
	(1144.15, -476.56, 34.97)	(1143.47, -477.43, 35.28)	(1143.61, -477.66, 34.54)	(1143.35, -478.36, 34.47)	(1143.43, -478.61, 35.62)		

calculated as follows [25]:

$$\begin{cases} \mu_{Cx} = \sqrt{{\mu_{cx}}^2 + {\mu_{rx}}^2} \\ \mu_{Cy} = \sqrt{{\mu_{cy}}^2 + {\mu_{ry}}^2} \\ \mu_{Cz} = \sqrt{{\mu_{cz}}^2 + {\mu_{rz}}^2}. \end{cases} \tag{20}$$

Therefore, the predicted results of the reference point (x_p, y_p, z_p) are calculated by the following equation:

$$\begin{cases} x_p = \dfrac{\sum_{i=1}^{n} x_{pi}}{n} \pm \mu_{Cx} \\ y_p = \dfrac{\sum_{i=1}^{n} y_{pi}}{n} \pm \mu_{Cy} \\ z_p = \dfrac{\sum_{i=1}^{n} z_{pi}}{n} \pm \mu_{Cz}. \end{cases} \tag{21}$$

The positioning error is calculated as follows:

$$\Delta e = \sqrt{(x_g - x_p)^2 + (y_g - y_p)^2 + (z_g - z_p)^2}. \tag{22}$$

An examination of the experimental results presented in Table VII shows that the combined standard uncertainty μ_{Cx}, μ_{Cy}, and μ_{Cz} is less than 0.6 mm, and the positioning error Δe is less than 0.4 mm. The experimental results prove that the stability and accuracy of the proposed method can meet industrial requirements.

D. Angle Iron Tracking Experiment With Continuously Changing Robot Pose

In order to test the calibration accuracy of the proposed method when the robot's pose changes continuously, angle iron was selected as the workpiece for our tracking experiment. We used the angle adjuster and the lip block to change the position and pose of the angle iron. Three groups of angle iron tracking experiments were carried out.

In each group of experiments, the pose of the robot at the tracking starting point and the tracking ending point changed significantly. The robot changed from the starting pose to the ending pose through interpolation, ensuring the substantial change of the robot's pose during the tracking process. Here, we took the first group as an example, as shown in Fig. 16. In the experiment, the camera collected images in real time and the robot controller calculated the ${}^{\mathbf{Base}}_{\mathbf{Wrist}}\mathbf{T}$. A series of images collected by the camera during the tracking process of the first group of experiments is shown in Fig. 17.

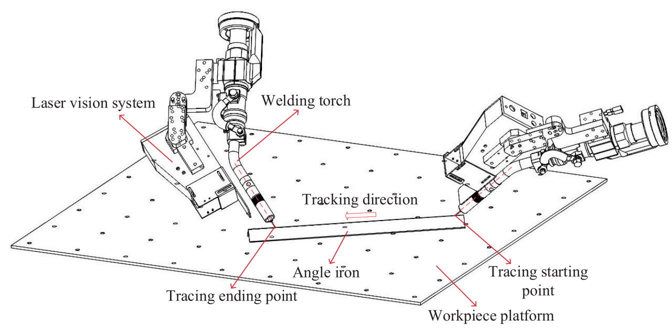

Fig. 16. Continuously changing robot pose tracking experiment of group 1.

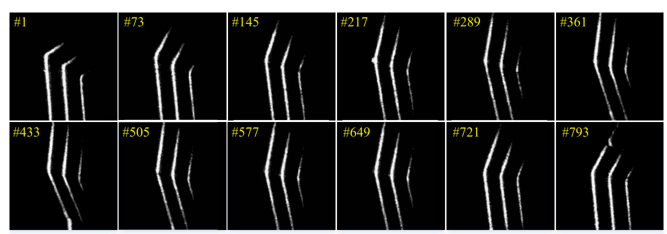

Fig. 17. Angle iron images of group 1. Note: #N in the top-left corner of the image means the N_{th} frame.

Comparative experiments were carried out with [8], [10]–[12], and [14] in terms of accuracy and stability. The hand–eye calibration conversion matrix ${}^{\mathbf{Wrist}}_{\mathbf{Laser}}\mathbf{T}$ of [8] and [10]–[12] was calculated through 279 calibration images in Section III. For [14], values of the robot wrist coordinate system were obtained by the input image collected by the

TABLE VIII
POSITIONING ERRORS OF THE ANGLE IRON

Group number	M1 MAE (mm)	M1 SD (mm)	M2 MAE (mm)	M2 SD (mm)	M3 MAE (mm)	M3 SD (mm)	M4 MAE (mm)	M4 SD (mm)	M5 MAE (mm)	M5 SD (mm)	M6 MAE (mm)	M6 SD (mm)
1	0.37	0.25	2.42	1.48	2.94	1.94	5.41	3.07	1.05	0.46	0.84	0.71
2	0.47	0.13	2.93	1.4	2.25	1.1	5.51	3.31	1.93	1.39	1.83	1.28
3	0.57	0.21	2.18	1.33	2.75	1.77	4.86	3.04	1.73	0.99	0.83	0.39

Note: M1, M2, M3, M4, M5 and M6 denotes our method and the method of [8], [10], [11], [12] and [14], respectively.

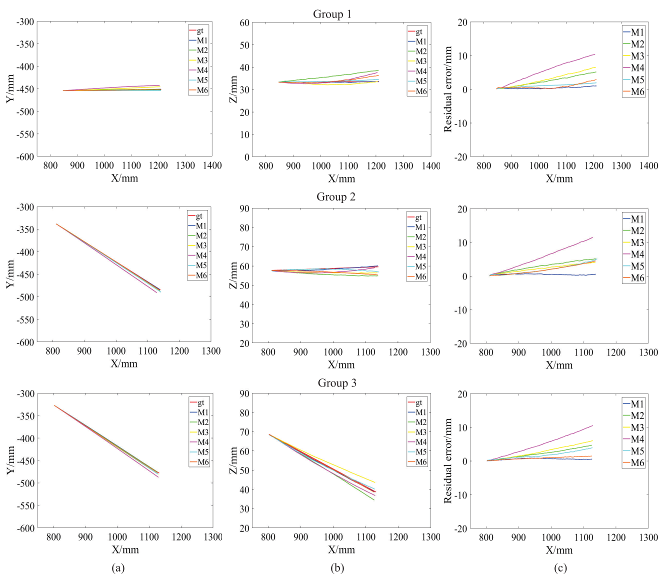

Fig. 18. Results of angle iron tracking experiment. (a) Tracking residual error on the plane of XOY. (b) Tracking residual error on the plane of XOZ. (c) Tracking residual error.

camera. The point morphology method of Section V-B was used to extract the pixel coordinates of the feature points and input them into the GAN model. Then, the GAN model outputs the wrist coordinates of the feature points. The tracking trajectories were obtained by converting wrist coordinates into robot base coordinates. Prior to each tracking experiment, the theoretical trajectory of angle iron was obtained by the teaching method. Based on (17), the Z-axis deviation was added and used to calculate the error between theoretical trajectory and tracking trajectory. The tracking errors are shown in Table VIII. Fig. 18 shows the tracking errors and tracking trajectories on the XOY and XOZ planes of three groups of experiments.

The maximum width of laser stripe is 10 pixels. According to the laser vision system parameters in Table IV, 10 pixels are approximately equal to 1.25 mm. It is observed from Table III and Fig. 18 that when the robot's pose changes greatly, the MAE values of the M2, M3, M4, and M5 methods are greater than mm, which cannot meet the requirements. M6 satisfies the accuracy requirement under groups 1 and 3, but its SD is still large. By contrast, the MAE of our proposed algorithm M1 is within 0.6 mm and the SD is within in 0.3 mm, showing good accuracy and stability.

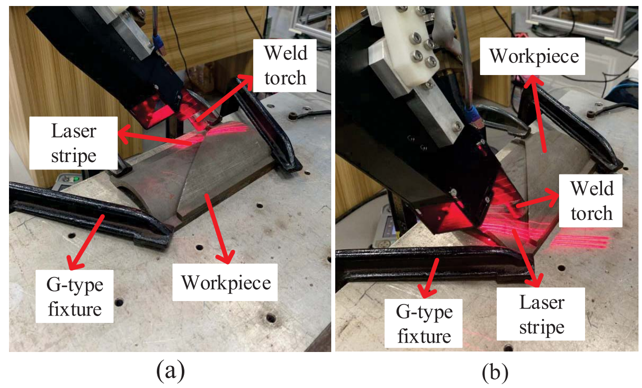

Fig. 19. Poses of tracking starting and ending point of the 3-D-L/I. (a) Pose of the starting point. (b) Pose of the ending point.

E. Welding Seam Tracking Experiment of Complex 2-D and 3-D Curve

To verify the accuracy of the proposed method under the environment of tracking complex 2-D and 3-D curves, six types of welding workpieces are used for the weld tracking experiments. In terms of tracking methods, this part is similar to Section V-C, with only different poses for the tracking starting point and the ending point. In the weld tracking experiment, the tracking starting point and ending point poses are determined by the shape of the weld. We use 3-D-L/I for presentation, as shown in Fig. 19.

In the calculation of errors, the tracking error calculation is directly based on (17) because the 2-D welding workpiece only considers the tracking errors on the XOY plane. The tracking error calculation of the 3-D welding workpiece is the same as that of the angle iron workpiece in Section V-C. The tracking errors are shown in Table IX. Fig. 20 shows the tracking errors and tracking trajectories on the XOY planes of 2-D-B/C, 2-D-B/S, and 2-D-L/C. Fig. 21 shows the tracking errors and tracking trajectories on the XOY and XOZ planes of 3-D-B/C, 3-D-B/I, and 3-D-L/I.

It can be observed from Table X and Figs. 20 and 21 that the methods M2–M5 do not perform well in tracking various workpieces. M6 has a good tracking effect in the 2-D welding workpieces. However, its calibration data are 2-D and lack of diversity and its tracking effect in 3-D welding workpieces is poor, so it cannot meet the requirements of tracking accuracy and stability. By contrast, the MAE and the SD of our method M1 are both within 0.4 mm, showing good tracking accuracy and stability.

TABLE IX

POSITIONING ERRORS OF THE WELDING WORKPIECES

Workpieces types	Group number	M1 MAE (mm)	M1 SD (mm)	M2 MAE (mm)	M2 SD (mm)	M3 MAE (mm)	M3 SD (mm)	M4 MAE (mm)	M4 SD (mm)	M5 MAE (mm)	M5 SD (mm)	M6 MAE (mm)	M6 SD (mm)
2D-B/C	1	0.34	0.18	0.57	0.28	0.62	0.29	0.6	0.33	0.44	0.27	0.46	0.29
2D-B/S	2	0.29	0.12	1.19	0.97	1.28	0.9	0.58	0.47	1.36	0.94	0.4	0.15
2D-L/C	3	0.27	0.18	0.67	0.35	1.56	1.34	1.23	0.41	1.3	1.19	0.44	0.26
3D-B/C	4	0.32	0.31	1.42	1.17	1.05	0.85	2.87	1.17	2.19	0.48	1.44	0.78
3D-B/I	5	0.35	0.36	1.36	0.66	1.78	1.19	4.88	2.36	1.47	0.97	1.19	1.34
3D-L/I	6	0.38	0.29	1.86	1.45	1.31	0.76	3.45	2.47	1.51	1.76	1.06	0.68

Note: M1, M2, M3, M4, M5 and M6 denotes our method and the method of [8], [10], [11], [12] and [14], respectively.

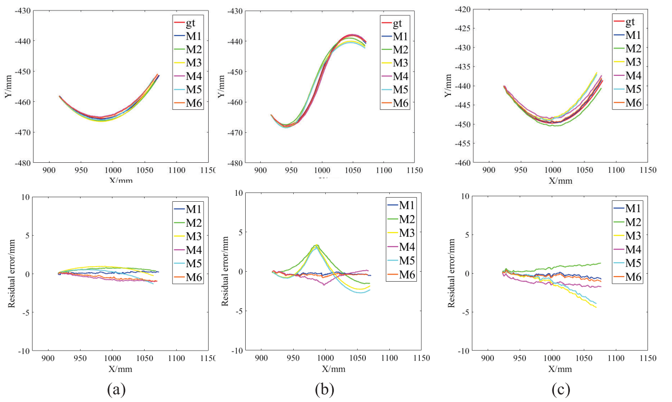

Fig. 20. Trajectory tracking results and tracking residual error of 2-D welding workpieces tracking experiment. (a) 2-D-B/C. (b) 2-D-B/S. (c) 2-D-L/C.

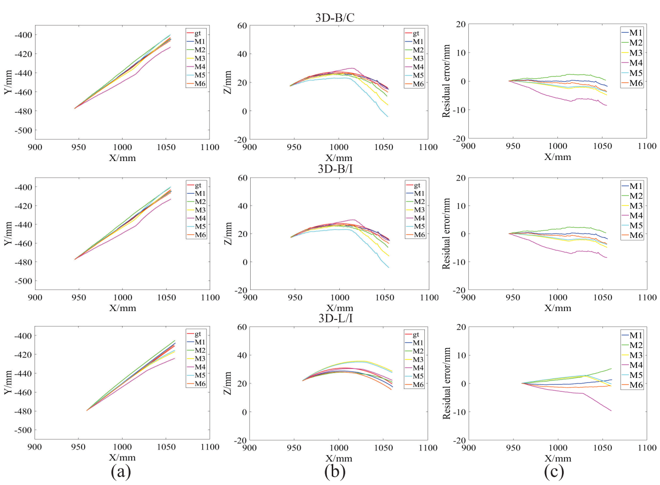

Fig. 21. Results of 3-D welding workpieces tracking experiment. (a) Tracking residual error on the plane of XOY. (b) Tracking residual error on the plane of XOZ. (c) Tracking residual error.

VI. CONCLUSION

The following conditions hold.

1) To improve the calibration efficiency and accuracy, a calibration optimization method for the welding robot laser vision system based on the GAN was proposed in this article. By establishing an accurate calibration model, the calibration process is transformed into an optimization problem for the calibration parameters. We used the calibration data to optimize the calibration parameters with the GAN. After that, a precise conversion from the pixel coordinate system to the robot base coordinate system was obtained.
2) To provide accurate, diverse, and sufficient calibration data for the calibration optimization method, we proposed an intelligent collection method that collects a large amount of unlabeled calibration data and a small amount of labeled calibration data efficiently and intelligently.
3) We compared our intelligent collection method with the traditional one. Our method took 1/3 time to collect about 61.5 times as many images without too much manual participation. It significantly improved the calibration efficiency and provides accurate, diverse, and sufficient calibration data sources for the calibration method.
4) The reference point positioning experiment, angle iron tracking experiment with continuously changing robot pose, and welding tracking experiment with complex 2-D and 3-D curves were carried out. In the reference point positioning experiment, the combined standard uncertainty μ_{CX}, μ_{CY}, and μ_{CZ} was less than 0.6 mm and the positioning error Δe was less than 0.4 mm, showing that our method had low uncertainty and good stability in point positioning. In the angle iron tracking experiment, the MAE and SD were controlled within 0.6 and 0.3 mm, respectively. The experimental results demonstrated that our method still had good calibration accuracy and stability in the continuously changing robot pose. In the welding tracking experiment, the MAE and the SD are both within 0.4 mm, illustrating that our method can track complex and changeable curves. The above three experiments proved the effectiveness and robustness of the proposed method, which can well meet the industrial requirements.

REFERENCES

[1] J. S. Kim, Y. T. Son, H. S. Cho, and K. I. Koh, "A robust visual seam tracking system for robotic arc welding," *Mechatronics*, vol. 6, no. 2, pp. 141–163, 1996.

[2] D. Xu, M. Tan, X. Zhao, and Z. Tu, "Seam tracking and visual control for robotic arc welding based on structured light stereovision," *Int. J. Autom. Comput.*, vol. 1, no. 1, pp. 63–75, 2004.

[3] X. Li, X. Li, S. S. Ge, M. O. Khyam, and C. Luo, "Automatic welding seam tracking and identification," *IEEE Trans. Ind. Electron.*, vol. 64, no. 9, pp. 7261–7271, Sep. 2017.

[4] Y. Tian, W. R. Hamel, and J. Tan, "Accurate human navigation using wearable monocular visual and inertial sensors," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 1, pp. 203–213, Jan. 2014.
[5] A. M. McIvor, "Calibration of a laser stripe profiler," in *Proc. 2nd Int. Conf. 3-D Digit. Imag. Modeling*, 1999, pp. 92–98.
[6] S. Zhang and P. S. Huang, "Novel method for structured light system calibration," *Opt. Eng.*, vol. 45, no. 8, Aug. 2006, Art. no. 083601.
[7] Y. C. Shiu and S. Ahmad, "Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form $AX = XB$," *IEEE Trans. Robot. Autom.*, vol. 5, no. 1, pp. 16–29, Feb. 1989.
[8] R. Y. Tsai and R. K. Lenz, "A new technique for fully autonomous and efficient 3D robotics hand/eye calibration," *IEEE Trans. Robot. Autom.*, vol. 5, no. 3, pp. 345–358, Jun. 1989.
[9] K. Daniilidis, "Hand-eye calibration using dual quaternions," *Int. J. Robot. Res.*, vol. 18, no. 3, pp. 286–298, 1999.
[10] F. C. Park and B. J. Martin, "Robot sensor calibration: Solving $AX = XB$ on the Euclidean group," *IEEE Trans. Robot. Autom.*, vol. 10, no. 5, pp. 717–721, Oct. 1994.
[11] H. Nguyen and Q.-C. Pham, "On the covariance of X in $AX = XB$," *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1651–1658, Dec. 2018.
[12] J. Wu, Y. Sun, M. Wang, and M. Liu, "Hand-eye calibration: 4-D procrustes analysis approach," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 2966–2981, Jun. 2020.
[13] Y. Zou and X. Chen, "Hand–eye calibration of arc welding robot and laser vision sensor through semidefinite programming," *Ind. Robot, Int. J.*, vol. 45, no. 5, pp. 597–610, Dec. 2018.
[14] Y. Zou and R. Lan, "An end-to-end calibration method for welding robot laser vision systems with deep reinforcement learning," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 7, pp. 4270–4280, Jul. 2020.
[15] C. J. Merz, D. C. S. Clair, and W. E. Bond, "Semi-supervised adaptive resonance theory (SMART2)," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, vol. 3, 1992, pp. 851–856.
[16] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 1–9.
[17] A. Odena, "Semi-supervised learning with generative adversarial networks," 2016, *arXiv:1606.01583*. [Online]. Available: http://arxiv.org/abs/1606.01583
[18] M. Ding, J. Tang, and J. Zhang, "Semi-supervised learning on graphs with generative adversarial nets," in *Proc. 27th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2018, pp. 913–922.
[19] Y. Wei *et al.*, "Semi-DerainGAN: A new semi-supervised single image deraining," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2021, pp. 1–6.
[20] Y. Zou, J. Chen, and X. Wei, "Research on a real-time pose estimation method for a seam tracking system," *Opt. Lasers Eng.*, vol. 127, Apr. 2020, Art. no. 105947.
[21] J. Heikkila, "Geometric camera calibration using circular control points," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 10, pp. 1066–1077, Oct. 2000.
[22] Y. Zhang, Z. Gan, and L. Carin, "Generating text via adversarial training," in *Proc. NIPS Workshop Adversarial Training*, vol. 21, 2016, pp. 21–32.
[23] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.
[24] Y. Zou and W. Zhou, "Automatic seam detection and tracking system for robots based on laser vision," *Mechatronics*, vol. 63, Nov. 2019, Art. no. 102261.
[25] *Guide to the Expression of Uncertainty in Measurement*, ISO, OIML, Geneva, Switzerland, 1995, pp. 16–17, vol. 122.

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